



Identify user features impacting keystroke, mouse and touchscreen dynamics

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Abstract

To understand how individual user features affect one's emotional changes in response to stimuli or changes in a situated external environment, it is necessary to measure how user features affect one's behavioral changes (patterns of changes) in the use of keyboard, mouse, and touchscreen devices (KMT dynamics), as the measurable behavioral changes are related to both user features and the user's emotional states. Our central research questions are: (1) which user features have a significant impact on behavior such as KMT dynamics? (2) how do these user features impact behavior from qualitative and quantitative perspectives? and (3) what is an effective framework to support personalized emotion recognition? To answer these questions, we conducted a questionnaire survey with 107 participants to investigate how, and to what extent, user profile features can influence keyboard, mouse, and touchscreen usage in dynamics terms (KMT dynamics). We then performed data analysis to explore the relationships between an initial set of user feature candidates and an initial set of KMT dynamics feature candidates (12 identified from a literature review). Based on the identified relationships, we identified seven important user features and seven important KMT dynamics features for further KMT-based emotion recognition studies. The seven key user features are: gender, KMT proficiency, language proficiency, hand using habit, software proficiency, task urgency, and emotion, while the seven important KMT dynamics features are: typing speed, typing error rate, pressure value of key/strength of pressing touchscreen, mouse (finger or pencil) movement speed, clicking speed, clicking strength, and keystroke speed. Finally, we proposed a new emotion recognition framework incorporating the identified key user and KMT dynamics features as a guiding model for further research in this field.

Keywords User profile · KMT dynamics features · Feature streamlining · Personalized emotion recognition framework · Qualitative analysis · Quantitative analysis

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1 Introduction

Behavior and emotion are two different forms of human reactions to external environments or stimuli [1, 2]. Behavior is more of a physical manifestation, while emotion is more inclined to internal cognitive, motivational and subjective elements [3, 4]. The main difference between these two forms is that behavior can be more easily measured using technology [5–8], whereas emotion is difficult to assess directly and is typically inferred from changes in a subject's behavior, such as facial expressions [9, 10], body gestures [11], eye movements [12], voice and tone [13], physiological signals [14], and keystroke, mouse, and touchscreen (KMT) dynamics [15]. Thus, instead of measuring emotions directly, assessing behavioral changes to infer emotions is a practical and effective approach in the field of affective computing [16, 17]. User features, such as gender, age, education level, handedness, and language could introduce bias in emotion recognition based on various indicator signals [18–20]. However, most existing research focuses on recognizing emotions from various behavioral changes without considering individual user features, which limits their ability to provide personalized emotion recognition for individual well-being.

To develop personalized emotion recognition, here we propose an initial framework that incorporates user features into the emotion recognition process (see Fig. 1). This framework is partially derived from our previous literature review work [15] and is grounded in the established relationship between the mind and behavior [1]. It is well observed that, even under the same situation, individuals may exhibit different behavioral and emotional responses [21], which are associated with individual differences [22, 23]. That means user

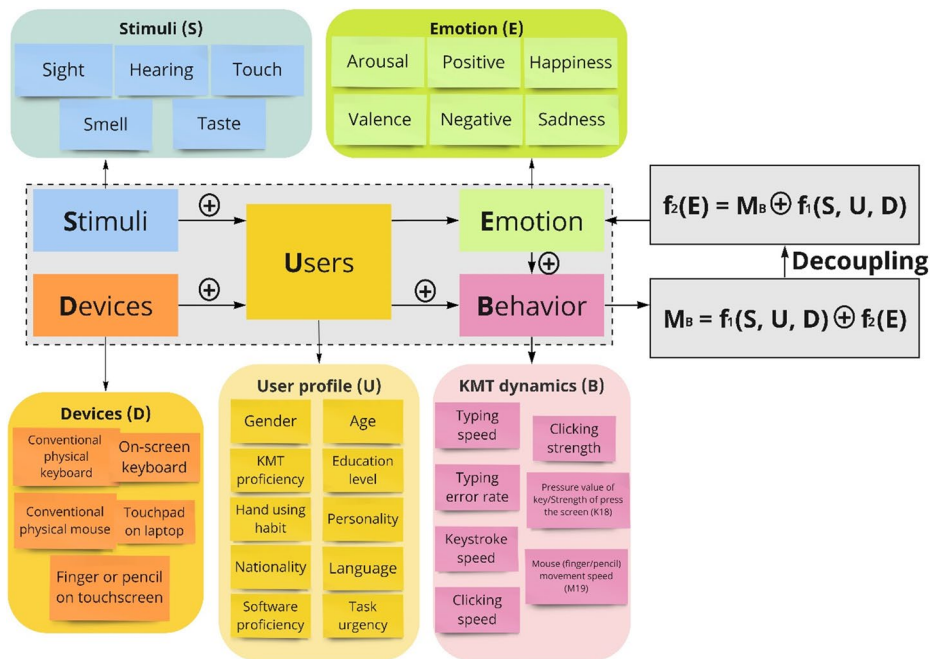


Fig. 1 The initial personalized emotion recognition framework

features are the users' underlaid/embedded factors affecting their behavior and emotion simultaneously [24]. A set of a user's features is collectively referred as the user profile.

We model the relationships among Stimuli (S), Emotion (E), Device (D), User profile (U), and Behavior (B) in Fig. 1, which serves as a guiding framework for personalized emotion recognition. The measurement of behavior (M_B) is coupled with stimuli (S) [25–29], user profile (U) [21], the device used (D) (keyboard, mouse and touchscreen (KMT) devices) [22, 30], and emotion (E) [2]. Their relationships are defined in Eq. (1), and the corresponding inverse problem is described in Eq. (2).

$$M_B = f_1(S, U, D) \oplus f_2(E) \quad (1)$$

$$f_2(E) = M_B \oplus f_1(S, U, D) \quad (2)$$

Equation (1) describes the factors that affect behavior, while Eq. (2) explains how to interpret emotional states from the measured behavior. The symbol $\oplus$ represents a coupling relationship between two interrelated elements. Thus, to predict emotions more accurately from behavior changes, personal differences or user features need to be taken into account as well. However, it still remains unclear how to incorporate user features into behavioral features for emotion recognition [15]. To address this question, three sub-questions need to be investigated: (1) which user features have a significant impact on behavior such as KMT dynamics, (2) how do these user features impact behavior from qualitative and quantitative perspectives, and (3) how the initial framework can be refined based on the answers to the above questions to guide future research.

Referring to Fig. 1, given a set of measurements on KMT dynamics, some user features can be predicted if the emotion is not involved. This can be seen as decoupling the simplified Eq. (1) $M_B = f_1(S, U, D)$. For example, recent studies [31–34] have successfully predicted the user's age [32], gender [31, 32], and personality [33, 34] based on some measurements on KMT dynamics. These findings suggest that a user's personal profile has impact on the user's KMT behavior and in reverse analysing KMT dynamics can interpret the user profile [35].

Note that this initial framework focuses only on behavior changes related to KMT dynamics. The reasons for this focus are as follows. First, facial expressions, body gestures, and eye-tracking are external bodily reactions, which are easily observed but can sometimes be deceptive [9, 11–13]. Second, physiological signals such as EEG, heart rate, and blood pressure are generally more authentic but hard to capture [14]. KMT dynamics refer to how individuals interact with KMT devices in their daily life and work [30, 36], and can be easily captured via applications or tools running in the background without interrupting users' workflow. Thus, the non-intrusive and unobtrusive nature of KMT dynamics makes them a better choice to predict a user's emotional state while using KMT devices [15, 36]. Given this, the present study specifically investigates how individual elementary user features impact on KMT dynamics.

Regarding to our first research question, previous studies have demonstrated that several user features such as gender, age, handedness, education level, and personality have coupled relationships with KMT dynamics [22, 37, 38]. Researchers have also successfully predicted these user features from measuring KMT dynamics (patterns of changes) with a relatively high accuracy [31, 33, 34, 39–42]. However, additional user features, includ-

ing cultural context, proficiency with KMT devices, task-related software proficiency, task urgency, and emotional states, are also likely to influence a user's KMT dynamics. Thus, there is still considerable scope to investigate which user features significantly influence KMT dynamics and to what extent these key user features affect KMT dynamics.

Specifically, 12 commonly used KMT dynamics features summarised in [15] are studied against user features in this study. The abbreviations of these KMT features are listed below:

- Typing speed (K6)- Total number of words typed in a certain time unit.
- Typing error rate (K7)- Frequency of delete or backspace keys can reflect error rate/correction rate.
- Pressure value of Key/Strength of pressing touchscreen (K18)- The strength of pressing a key/screen.
- Mouse (finger or pencil) movement speed (M19)- The mouse speed in each movement.
- Mouse clicking speed (F1)- The time from a button down to the same button up.
- Mouse clicking strength (F2)- The strength of pressing a button.
- Keystroke speed (F3)- The time from a key down to the same key up.
- Latency time (K2)- The time from a key release to a key down behavior.
- Duration of the same key (K1)- Interval between key press and release of the same key (1D) or two consecutive keys (digraphs or 2D) or three consecutive keys (trigraphs or 3D).
- Down-down time of two consecutive keys (K3)- The time from a key down to another key down.
- Total keys pressed (K21)- The number of all key events.

This study addresses the research questions by conducting a questionnaire survey with 107 participants to qualitatively identify both important user and KMT dynamics features. It further studies the extent to which the identified key user features affect certain KMT dynamics features. Based on the findings, the initial personalized emotion recognition framework is updated by incorporating the identified key user and KMT dynamics features, providing guidance for future research. The main contributions of this study are as follows:

1. Identification of key user features that significantly influence the behavior related to KMT devices usage.
2. Identification of key KMT dynamics features as important behavioral indicators for emotion recognition.
3. Understanding of how these user features affect the KMT dynamics features from both qualitative and quantitative perspectives.
4. A new personalized emotion recognition framework is proposed to integrate the identified key user and KMT dynamics features into the emotion recognition pipeline.

The remaining parts of this paper are organized as follows. Section 2 reviews related work and identifies research gaps; Sect. 3 describes our methodology, which includes both qualitative study and quantitative study; Sect. 4 presents a detailed analysis and the corresponding results; Sect. 5 discusses key research findings, potential future implementations of the proposed framework, and some limitations of this study; and Sect. 6 concludes this study.

2 Related work and research gaps

2.1 Related work on relationship between user features and KMT dynamics features

2.1.1 Gender in relation with KMT dynamics

Gender is the most studied user feature in relation to KMT dynamics. It has been investigated that males and females exhibit different behavior and have significantly different ways of expressing emotions [24]. The related work is summarized in Table 1, involving devices, KMT dynamics, and gender distinction.

The research in [31] focused on predicting gender, based on average values such as the duration of the same key (K1), down-down time between two consecutive keys (K3), and typing speed (K6) on a physical keyboard. The study found that typing speed (K6) alone is insufficient for accurate gender recognition. Similarly, the research in [42] investigated gender-based differences in typing behavior, including typing speed (K6), latency time (K2), density of backspaces (K7), backspace time, chunk length, chunk duration, and density of suspension points. These gender-based typing behavior differences were found to be consistent across various KMT devices. The differences show that females are more likely to type faster with fewer errors, and have shorter latency time, chunk duration, and fewer suspension points. The study in [43] identified five mobile typing dynamics features most useful for gender classification, which are: edit distance (between user input and displayed text), Gyrometer X coordinates, number of backspaces (K7), total keys pressed (K21), and Gyrometer Y coordinates. The finding supports the results of [42] that males tend to make more typing errors than females.

Several studies [35, 44–46] have also explored gender recognition using KMT dynamics features. However, they did not provide a detailed analysis of which specific features are most effective in distinguishing between male and female users. In contrast to these findings, the research in [32] concluded that no features are significantly useful for gender prediction. These inconsistencies highlight the need for further research to determine whether KMT dynamics features can be reliably used for gender prediction, and if so, how to identify the most relevant features for this purpose.

Table 1 Relationships of gender and some KMT dynamics from existed research

Devices	KMT dynamics	Gender distinction
Conventional physical keyboard [31]	Typing speed (K6) [31, 42]	Female: faster↑ [42]
No restriction [42]	Latency time (K2) (Median) [42]	Female: shorter↓ [42]
Mobile phone [43]	Chunk length and duration [42]	Female: shorter↓ [42]
	Suspension points (density) [42]	Female: less↓ [42]
	Backspaces (K7) (time and density) [42, 43]	Female: less errors↓ [42, 43]

2.1.2 Age in relation with KMT dynamics

Age is another most widely studied user feature in relation to KMT dynamics. Table 2 summarizes the relevant studies that focus on age predication using KMT dynamics. These studies include both qualitative approaches [32, 43] and quantitative analysis [39].

Table 2 shows that typing speed (K6) is regarded as the most important keystroke dynamics for age detection [32, 43]. Young individuals aged 16 to 29 tend to type faster than other age groups on both virtual and physical keyboards [32]. In contrast, people over the age of 30 generally type more slowly, often using only one finger on touchscreen [43]. Teenagers aged 10 to 15 may also exhibit slower typing speeds, due to limited experience and skill development [32]. A study [39] found a linear relationship, showing that individuals around 20 years old type 62% faster and have 61% shorter pause times compared to those over 70.

Several other studies have identified a user's age through KMT dynamics [35, 44–47]; however, they provide limited information on which KMT dynamics features are most distinctive or how different age groups perform differently.

2.1.3 Other user features in relation with KMT dynamics

Some studies have examined other user features such as handedness, education level, number of hands (fingers) used, and personality. A consistent relationship has been observed in studies [41, 48] that individuals tend to (a) type faster on both touchscreen and physical keyboards when typing hand matches their handedness, and (b) type more slowly and less accurately with their non-dominant hand. Features identified as effective for handedness classification include the downtimes of each row's left and right of the keyboard [41], the duration of a single key (K1) and the down-down time between two consecutive keys (K3) [44]. However, in other studies, there is still no consensus on which KMT dynamics features offer the best discriminatory power for handedness identification [45, 49, 50]. Although the study [51] used user features such as age and handedness to recognize keystroke patterns and demonstrated that different user profiles lead to distinct keystroke patterns, further research is needed to explore these relationships in depth and to provide a more solid foundation in this area.

Education level has also been examined as a user feature in studies [40, 44, 52]. Time related keystroke features, whether from touchscreen or physical keyboards, are commonly used to interpret differences in education levels across user groups [40, 44]. However, the influence of education level on other KMT dynamics features remains unclear. A similar uncertainty exists in identifying the number of hands (fingers) used during typing [43, 45, 46]. The study in [43] found that some time related features and gyroscope Y and Z coordi-

Table 2 Relationships between age and some KMT dynamics from related research

Devices	KMT dynamics	Age distinction
No restriction	Typing speed (K6) [32]	16–29 years old: higher
	(Pause time between	speed↑ [32]
Touchscreen	words) [43]	(Less pause ↓ [43])
[39, 43]	Timestamps of keypress	A positive linear effect for
	events and their type	age: younger users (around
	[39]	20 years old) typed 62%
		faster and paused 61% less
		than older users (above 70
		years old). [39]

nates are particularly useful for detecting the number of fingers used while typing on mobile phones. In contrast, other researchers [45] have argued that further investigation is needed to determine the effectiveness of gyroscope and accelerometer features.

Recognition of personality traits based on various KMT behavior has also been explored [33, 34]. Recently, a study [53] explored personality prediction using hand-tracking and keystroke data during typing tasks in Augmented Virtuality and Virtual Reality. The results demonstrated that certain keystroke features such as Duration of the same key (K1) and Latency time (K2), can be effectively used to predict personality traits. However, the influence of personality traits on KMT behavior remains unclear.

2.2 Research gaps

We summarized the interrelationships among devices, commonly used KMT dynamics features, and recognized user features in Fig. 2. Most studies have identified one or more user features using keystroke dynamics [54], whereas mouse dynamics has been less frequently explored [32]. Although some studies have used mouse features to recognize user gender and age [20, 32], the impact of user features on mouse dynamics remains under-investigated. Further research is needed to explore this relationship in more depth.

Existing research has studied the correlations between various user features (gender, age, handedness, education level, number of hands (fingers) used) and commonly used KMT dynamics especially time related features, which are easier to record and measure [45]. In the context of education levels recognition, time related keystroke features are also frequently used, but it remains unclear which specific features are more effective. Thus, we did not connect education level to any particular feature in Fig. 2. Despite existing research efforts, understanding relationships between KMT dynamics and user features requires further investigation, as no consensus has been reached from either qualitative or quantitative perspectives.

Most notably, none of the aforementioned studies have examined emotion as a user feature. This clear gap highlights the need for further investigation in this area.

The recognition methods in reviewed research literature can be classified into statistical analyses, such as T-tests [42] and hierarchical growth curve mixed-effects models [39],

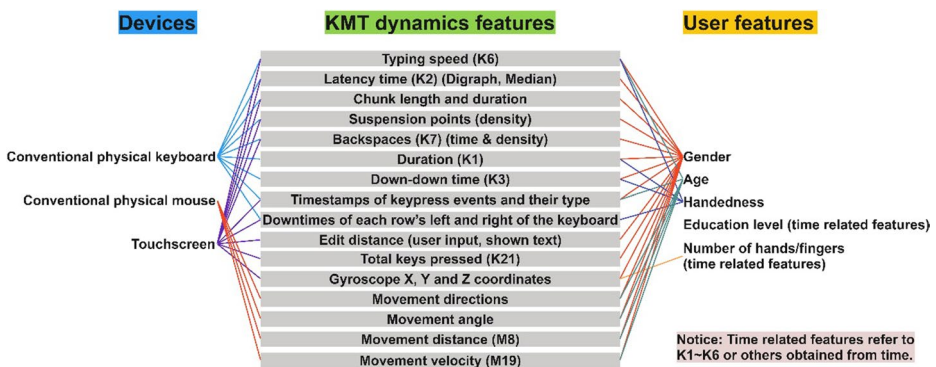


Fig. 2 Interrelationships among devices, KMT dynamics, and user features based on the reviewed research

and machine learning algorithms including Naïve Bayes (NB), Random Forest (RF), and K-Nearest Neighbour (KNN) [43–45, 52]. A keystroke dynamics dataset containing gender, age group, handedness, mother tongue, and education level has been built for supporting user features classification through AI models [19]. Most of these methods rely on black-box machine learning algorithms, which do not clearly reveal the relationships between KMT dynamics features and user features. This limitation motivated us to use a questionnaire survey with multi-choice questions to explicitly explore these relationships through human intelligence rather than relying solely on machine intelligence.

3 Research methodology

This study applies a questionnaire survey covering a broad range of user feature candidates and KMT dynamics feature candidates to gain a comprehensive understanding of which features are most important in both categories. A questionnaire survey with 107 participants was conducted to answer the above questions from both qualitative and quantitative perspectives. The main steps are (1) to determine what user feature candidates to be investigated; (2) to define what KMT dynamics feature candidates to be studied; (3) to design questionnaire and determine sample size; (4) to validate designed questionnaire with a pilot study; (5) to conduct the survey; and (6) to analyse data for drawing conclusions.

3.1 User feature candidates

Building on some previous research [39, 40, 42, 48] that focused on gender, age, handedness, education level, number of hands (fingers) used, and personality, and our previous review [15], user feature candidates to be studied in this survey are: (a) Gender, (b) Age, (c) Education level, (d) Personality, (e) Language proficiency, (f) Hand using habit when using a touchscreen, (g) KMT proficiency/familiarity, (h) Software proficiency, (i) Task urgency, and (j) Emotion.

KMT behavior differences across age groups and education levels are mainly caused by individuals' familiarity or proficiency with KMT devices [32, 43]. Therefore, we will analyse the correlations between age and education level with KMT device familiarity, rather than directly examining their relationships with KMT dynamics.

Emotion is also an important factor affecting KMT dynamics and can be considered as a user feature associated with a user. However, compared to the above user features, emotion is a more flexible factor and often changes because of many complex reasons. To make it easy to understand the decoupling of emotion with KMT dynamics and other user features, this study describes emotion as a kind of dynamic user feature rather than other static features listed from a) to i). And in order to signal this dynamic feature out, in Eqs. (1) and (2), emotion is separated from the user profile features.

3.2 KMT dynamics feature candidates

Similarly, from the related work mentioned in this paper and key literatures reviewed in [15], some significant KMT dynamics feature candidates are summarised as follows. Their

abbreviations are used as the same as in [15] for easy to refer to, such as K6, K7, K18, M19 come from [15] and F1, F2, F3 are found in this study.

- *Typing speed (K6).*
- *Typing error rate (K7).*
- *Pressure value of Key/Strength of pressing touchscreen (K18).*
- *Mouse (finger or pencil) movement speed (M19).*
- *Clicking speed (F1).*
- *Clicking strength (F2).*
- *Keystroke speed (F3).*

Another reason to choose these KMT dynamics features as candidates is that these features are easier for ordinary participants to understand and thus are more suitable for a questionnaire survey.

3.3 Questionnaire design

The questionnaire is to investigate whether the listed user feature candidates could impact the defined KMT dynamics feature candidates and how these user feature candidates affect KMT behavior. To understand the correlation between user feature candidates and KMT behavior, a total of 31 questions were included in this questionnaire and all questions were divided into basic information part asking for participants’ personal information and main body part asking for their opinions. Since some participants might not be familiar with KMT dynamics terminology and thus making it difficult for them to answer open questions, we set all questions with multiple choices with scales in plain language. Two examples are shown below:

Question 9 How often do you usually use the following keyboard and mouse in your study or work (multi choice)?

	Never	Very rarely	Rarely	Occasionally	Frequently	Very frequently
Conventional physical keyboard	☐	☐	☐	☐	☐	☐
On-screen keyboard	☐	☐	☐	☐	☐	☐
Conventional physical mouse	☐	☐	☐	☐	☐	☐
Touchpad on laptop	☐	☐	☐	☐	☐	☐
Finger or pencil on touchscreen	☐	☐	☐	☐	☐	☐

Question 24 When using on-screen keyboard (such as smart phone and pad) to type,what are the differences of two-hand typing compared with single hand typing (multi choice)?

	Higher/Larger	Lower/Smaller	No difference	Not clear
Typing speed:	☐	☐	☐	☐
Typing error rate:	☐	☐	☐	☐
Strength of press the screen:	☐	☐	☐	☐
Movement speed of finger or pencil:	☐	☐	☐	☐

3.4 Distribution and data collection

We launched our questionnaire on the free questionnaire editing and distribution platform <https://www.wjx.cn/>, the questionnaire was then distributed online to our research team, colleagues, and friends, without restrictions to any specific group. Within two weeks, we received 107 answers in total. This sample size was considered sufficient compared to other studies that used relatively small sample sizes [15].

3.5 Reliability check for answers

To ensure the credibility of the analysis results, we first filtered out any invalid responses before analysing the answers from participants. We reviewed all 107 responses and confirmed that every participant completed the questionnaire. Therefore, all responses were valid in this survey.

4 Data analysis

4.1 Participant profile

Participant profile information was collected in the first part of the questionnaire which aimed to collect participants' basic information related to user feature candidates. The overall profile of 107 participants is described in Table 3.

The 107 participants include 57 females and 50 males, indicating a balanced gender distribution. Age distribution is that 64 participants aged from 25 to 34 years old, and 29 participants aged from 18 to 24. There are totally 14 participants aged from 35 to 64 years old. Unfortunately, no participant is under 18 years old or over 65 years old. There is a large proportion of participants (83.18%) who have obtained master's or higher degree and 14.95% participants having their bachelor's degrees. Just a few participants (1.87%) do not have a bachelor's degree. Each participant has attained at least a high school level of education. Most participants come from China and some from the UK. Another three participants come from Egypt, India, and Pakistan respectively. Their native languages almost align with their nationalities except one person. A large proportion of participants (93.46%) have learned a second language. There are 37.38% of participants who think themselves extroverts, while 62.62% identify as introverts. Participants' occupations include student, educator, researcher, writer, programmer, etc. Students accounted for the largest percentage at 67.29%, followed by educators and researchers, accounting for 14.01% in total. The habits of using keyboard show that most participants use two hands typing on-screen keyboards and ten fingers typing on physical keyboards. But there are also some users who use only their right hand or two fingers, respectively.

The proficiency/familiarity of using KMT devices was assumed to influence user's KMT behavior, so we asked participants to self-evaluate their degree of familiarity into different proficiency levels in Question 9 (seen in subsection 3.3) and Question 12 (below). The distributions of all responses are shown in Figs. 3 and 4, respectively.

Question 12 How familiar are you with using keyboard and mouse? (The type of keyboards and mouse will not be limited unless otherwise specified)

☐ Extremely ☐ Very ☐ Moderately ☐ Slightly ☐ Not at all

Table 3 Profile of participants

Personal features Sub-classes		Number of participants and their corresponding proportion relative to the total
Gender	Female	57 (53.27%)
	Male	50 (46.73%)
Age group	Under 18 years old	0 (0%)
	18–24 years old	29 (27.1%)
	25–34 years old	64 (59.81%)
	35–44 years old	7 (6.54%)
	45–54 years old	4 (3.74%)
	55–64 years old	3 (2.8%)
	65 or above 65 years old	0 (0%)
Education level	Less than high school	0 (0%)
	High school	2 (1.87%)
	Trade school	0 (0%)
	Bachelor's degree	16 (14.95%)
	Master's degree	55 (51.40%)
Nationality/Native language	PhD. or higher	34 (31.78%)
	British/English	8/7 (7.48%/6.54%)
	China/Chinese	96/97 (89.72%/90.65%)
	Egyptian/Arabic	1/1 (0.93%/0.93%)
	Indian/Tamil	1/1 (0.93%/0.93%)
Learned second language(s)?	Pakistani/Urdu	1/1 (0.93%/0.93%)
	Yes	100 (93.46%)
Personality	No	7 (6.54%)
	Extroversion	40 (37.38%)
Occupation	Introversion	67 (62.62%)
	Student	72 (67.29%)
	Educator	11 (10.28%)
	Lecturer/Researcher	4 (3.73%)
	Writer	1 (0.93%)
	IT programmer	5 (4.67%)
	Designer	2 (1.87%)
	Occupational Health and safety manager	1 (0.93%)
	Institutional staff	3 (2.8%)
	Technologist	2 (1.87%)
	Engineer	2 (1.87%)
	Sales representative	1 (0.93%)
	Auditor	1 (0.93%)
	Nurse	1 (0.93%)
	Environmentalist	1 (0.93%)
Hand(s)/Finger(s) used (touchscreen)	Food worker	1 (0.93%)
	Left hand	0 (0%)
	Right hand	40 (37.38%)
Hand(s)/Finger(s) used (physical keyboard)	Both hands	67 (62.62%)
	Ten fingers	76 (71.03%)
	Two fingers	31 (28.97%)

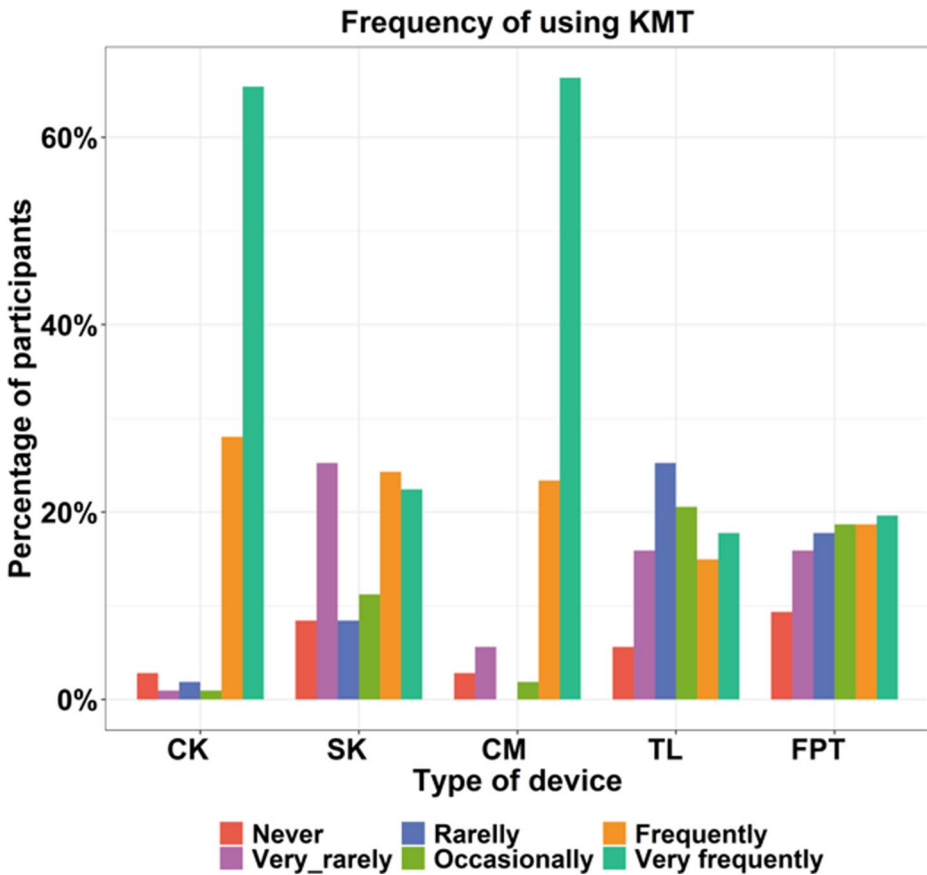


Fig. 3 Frequency of using KMT devices

CK, SK, and CM in Fig. 3 refer to the ‘Conventional physical keyboard’, ‘On-screen keyboard’, and ‘Conventional physical mouse’, respectively, while TL and FPT represent the ‘Touchpad on laptop’ and ‘Finger or pencil on touchscreen’, respectively. The y-axis indicates the percentage of participants at each frequency level relative to the total number of participants (this also applies to the y-axis in Fig. 4). The frequency of using KMT devices reveals that a large percentage (over 60%) of participants use conventional physical keyboards and mouse very frequently. Some participants also use the laptop touchpad and touchscreen at relatively higher frequencies. Although between 2.8% and 9.35% of participants reported never using one of these devices, none of them reported never using any type of keyboard and mouse, which to some degree reflects participants’ familiarity with at least one of these input devices, as illustrated in Fig. 4. Except for about 10% of participants who are moderately familiar with KMT devices, all others are at least very familiar with them. This suggests that participants’ experience of using KMT devices

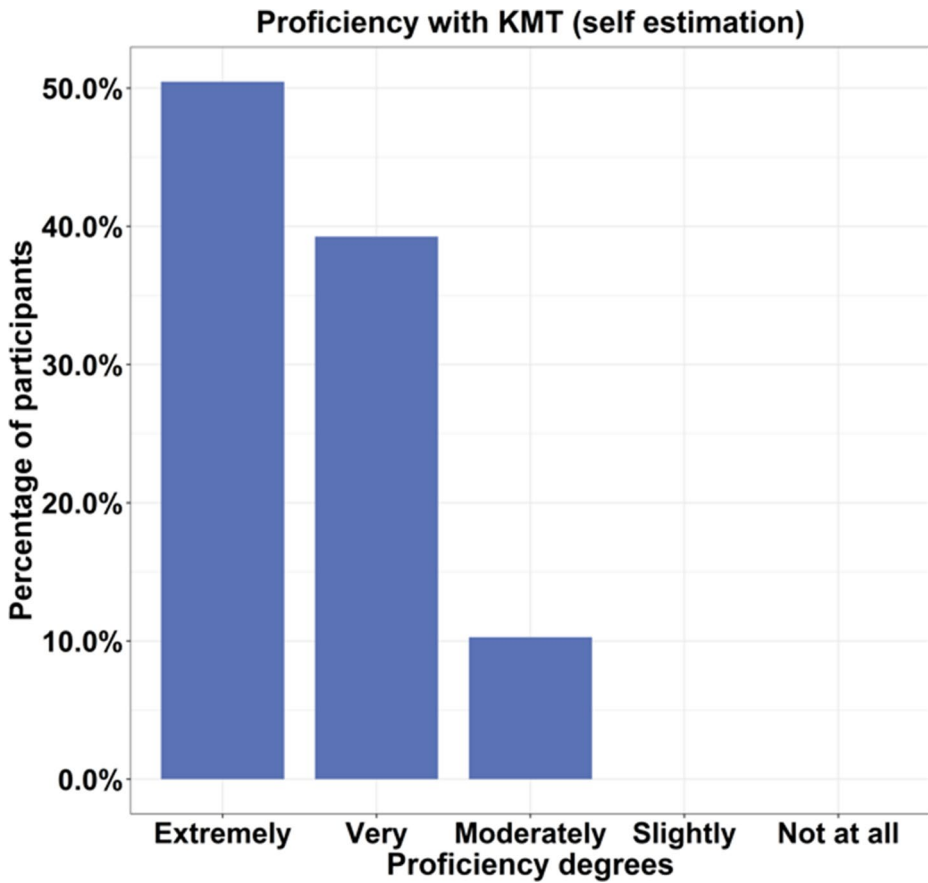


Fig. 4 Proficiency/familiarity with KMT devices

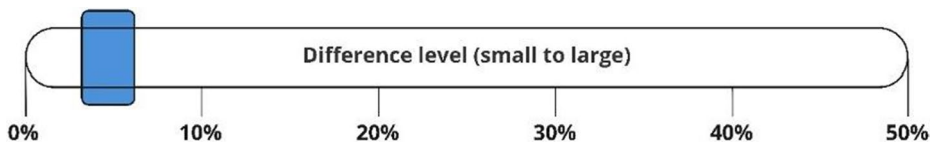
is rich and their perceptions of how user features affect KMT dynamics could be well-grounded in practical experience.

4.2 Qualitative analysis

To collect participants' opinions about the correlations between each user feature candidate and KMT dynamics feature candidate, we set two questions for each user feature candidate. The first question asked participants whether they believed a given user feature had any impact on KMT dynamics. If a participant believed there are some impacts, then they would be asked to answer the second question about the level of impact as a percentage. Otherwise, the second question was invisible for them. We set the maximum of the impact level to 50% because we believed a 50% impact from a single user feature is already a great impact level, regarding other features will make impacts too.

Question 13 Compared with female, what are the differences of male in the behavior of using keyboard and mouse?

	Higher/Larger	Lower/Smaller	No difference	Not clear
Keyboard				
Typing speed:	☐	☐	☐	☐
Typing error rate:	☐	☐	☐	☐
Pressure value of key:	☐	☐	☐	☐
Mouse				
Movement speed:	☐	☐	☐	☐
Clicking speed:	☐	☐	☐	☐
Clicking strength:	☐	☐	☐	☐



Question 14 How big is the difference between male and female in their keyboard and mouse behavior?

The participants' profile table (Table 3) shows that some user features such as age and education level, contain more than two sub-classes and the others have only two sub-classes. Table 4 summarizes the user features with only two sub-classes: gender is categorized as male or female, personality as extroversion or introversion, task urgency as urgent task or normal situation, and hand using habits as two-hand typing or single hand typing. Similarly, KMT proficiency, language proficiency, and software proficiency are classified as either more proficient or less proficient. Emotion is categorized as either agitation or soothing.

The relationships between each user feature candidate showed in Table 5 are described as four relative levels: higher/larger, lower/smaller, no difference, not clear between their two sub-classes. For example, male vs. female is for gender feature. Similarly, the other two user features against other sets of related KMT dynamics feature candidates are shown in Tables 6 and 7 respectively. The percentage figures in Table 5(a)-(f), Table 6, and Table 7 represent the proportion of participants (relative to the total number of participants) selecting each option regarding the perceived KMT performance trends under different user feature conditions. Thus, the sum of each row is about 100%. For example, in Table 5(a),

Table 4 Sub-classes of each user feature candidates

Gender	Male	KMT proficiency/familiarity	More proficient
	Female		Less proficient
Personality	Extroversion	Language proficiency	More proficient
	Introversion		Less proficient
Task urgency	Urgent task	Emotion	Agitation
	Normal situation		Soothing
Hand using habits in using keyboard	Two-hand typing	Software proficiency	More proficient
	Single hand typing		Less proficient

Table 5 KMT performance trends under different user feature conditions. (The figures in the table represent the percentage of participants who selected the corresponding options.)

(a) KMT performance trends by gender (male vs. female)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
K6	35.51%	8.41%	30.84%	25.23%
K7	19.63%	9.35%	34.58%	36.45%
K18	64.49%	4.67%	15.89%	14.95%
M19	49.53%	6.54%	21.50%	22.43%
F1	54.21%	3.74%	23.36%	18.69%
F2	57.01%	7.48%	19.63%	15.89%
(b) KMT performance trends by KMT proficiency (more proficient vs. less proficient)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
K6	81.31%	3.74%	11.21%	3.74%
K7	10.28%	71.96%	10.28%	7.48%
K18	32.71%	27.10%	23.36%	16.82%
M19	67.29%	5.61%	18.69%	8.41%
F1	64.49%	5.61%	19.63%	10.28%
F2	30.84%	21.50%	28.97%	18.69%
(c) KMT performance trends by personality (extroversion vs. introversion)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
K6	40.19%	10.28%	26.17%	23.36%
K7	31.78%	11.21%	27.10%	29.91%
K18	46.73%	4.67%	25.23%	23.36%
M19	41.12%	6.54%	29.91%	22.43%
F1	42.06%	9.35%	25.23%	23.36%
F2	40.19%	3.74%	32.71%	23.36%
(d) KMT performance trends by language proficiency (more proficient vs. less proficient)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
K6	71.03%	13.08%	14.02%	1.87%
K7	17.76%	64.49%	14.95%	2.80%
K18	29.91%	14.95%	48.60%	6.54%
M19	55.14%	11.21%	29.91%	3.74%
F1	56.07%	12.15%	26.17%	5.61%
F2	30.84%	13.08%	49.53%	6.54%
(e) KMT performance trends by task urgency (urgent task vs. normal situation)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
K6	85.05%	3.74%	11.21%	0.00%
K7	71.03%	11.21%	13.08%	4.67%
K18	73.83%	2.80%	21.50%	1.87%
M19	85.05%	2.80%	11.21%	0.93%
F1	80.37%	2.80%	15.89%	0.93%
F2	67.29%	4.67%	24.30%	3.74%
(f) KMT performance trends by emotions (agitation vs. soothing)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
K6	84.11%	8.41%	3.74%	3.74%
K7	74.77%	13.08%	8.41%	3.74%
K18	83.18%	7.48%	4.67%	4.67%
M19	80.37%	10.28%	4.67%	4.67%
F1	76.64%	10.28%	7.48%	5.61%
F2	80.37%	11.21%	3.74%	4.67%

Table 6 KMT performance trends of two-hand typing on touchscreen. (The figures in the table represent the percentage of participants who selected the corresponding options.)

KMT performance trends by hand using habit (two hands vs. one hand)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
K6	75.70%	10.28%	11.21%	2.80%
K7	26.17%	42.99%	24.30%	6.54%
K18	25.23%	34.58%	32.71%	7.48%
M19	65.42%	14.95%	13.08%	6.54%

Table 7 KMT performance trends for individuals who are more familiar with the software they use. (The figures in the table represent the percentage of participants who selected the corresponding options.)

KMT performance trends by software proficiency (more proficient vs. less proficient)				
Dynamics features	Higher/Larger	Lower/Smaller	No difference	Not clear
F3	82.24%	7.48%	8.41%	1.87%
K18	44.86%	25.23%	26.17%	3.74%
M19	85.98%	6.54%	5.61%	1.87%
F1	78.50%	10.28%	7.48%	3.74%
F2	38.32%	20.56%	37.38%	3.74%

“males vs. female” means that the trends are towards males, such as, 35.51% of participants thought that compared to females, males tend to have higher typing speed (K6).

Table 5 shows that, with the exception of personality, the other 5 user features are believed affecting KMT dynamics noticeably. Task urgency and emotion stand out as the most significant, with over 71.96% of participants acknowledging their influence (including both larger and lower). For example, in the case of emotion affecting K6, 84.11% of participants indicated a higher/larger speed and 8.41% for lower/smaller. The sum of them is 92.52%. When people perform an urgent task (for example the deadline is approaching) or do a normal task but under agitation emotional state, they tend to type, move mouse, and click mouse more quickly with larger pressure value and clicking strength. In these cases, a quicker speed may cause more typing errors. Personality shows no clear difference between the combined percentages of “higher/larger” and “lower/smaller” responses (50.47%-K6, 42.99%-K7, 51.40%-K18, 47.66%-M19, 51.41%-F1, 43.93%-F2) and the combined percentages of “no difference” and “no clear” responses (49.53%-K6, 57.01%-K7, 48.59%-K18, 52.34%-M19, 48.59%-F1, 56.07%-F2) across each KMT dynamics feature. Therefore, personality is considered not significantly affecting KMT dynamics.

Gender has a notable impact on the KMT dynamics features K18, M19, F1, and F2 between male and female users. Males tend to press key and mouse button more forcefully and move and click mouse more quickly than females. On the other hand, no significant differences in K6 and K7. This is inconsistent with findings presented in [31, 42, 43] (see Table 1), that males have lower typing speed since they make more typing errors and need to correct typos or misspellings more frequently. In this study, most participants think there is no difference in error rate between male and female, so, in this case, further investigation is necessary to study whether typing speed and error rate are significantly different between male and female.

Language proficiency shows significant differences in K6, K7, M19, and F1 between two situations where people are more proficient and less proficient with a language. This finding is more credible given that 93.46% of participants (Table 3) have learnt at least one second language, and over 85% report that they perform much poorer (~60%) or a little bit poorer (~25%) when using a second language compared to their native language (see Fig. 5). Therefore, participants can give answers to this question based on their own experience. When using a more proficient language, people are more likely to type faster with fewer errors and move and click the mouse more quickly. This finding is understandable with our common sense.

KMT proficiency significantly differentiates all KMT dynamics except F2 between participants who are more proficient with KMT devices and those who are less proficient. Individuals who use KMT devices more frequently in their daily work and life tend to be more familiar with them. It can be found that the familiarity with KMT devices shows a positive correlation with K6, M19, and F1, and a negative correlation with K7, consistent with findings in [32, 43]. That is to say that people who are more proficient with KMT devices tend to type faster with fewer errors and move and click the mouse more quickly compared to less proficient users. Some differences in K18 also appear among participants with varying proficiency levels, but it is hard to say whether these differences are positive

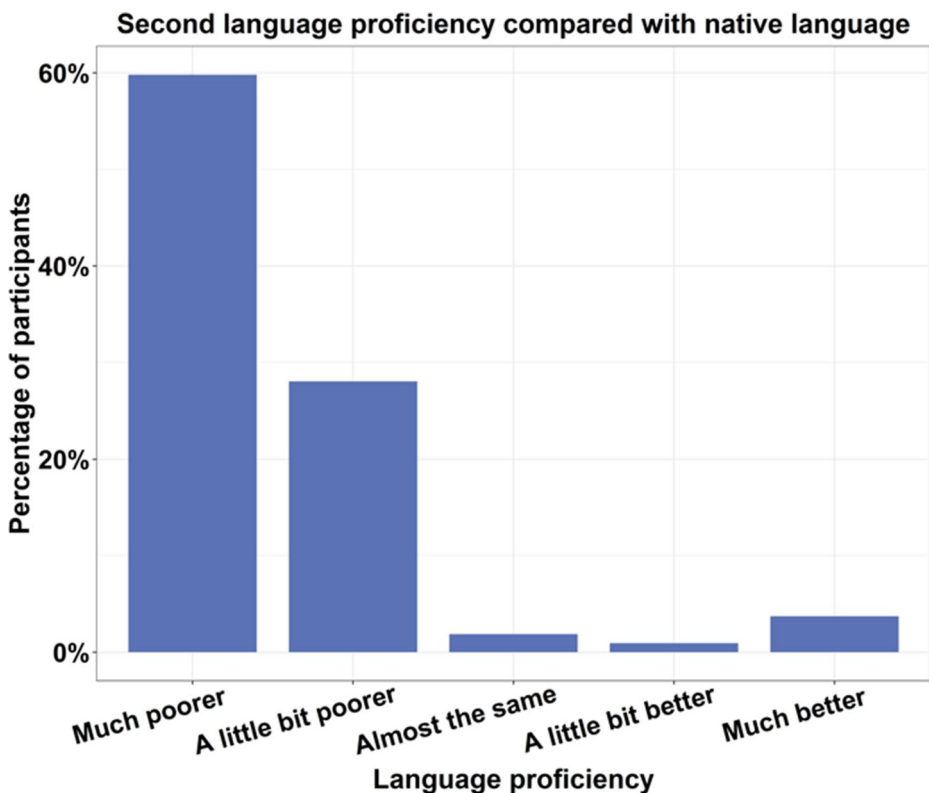


Fig. 5 Participants distribution in terms of the proficiency of second language compared with native language

or negative because nearly equal percentages of participants hold opposing opinions. Thus, more research should be conducted to make sure what the differences mean.

The review of related work in Sect. 2 showed that age and education levels influence on KMT dynamics mainly because different age groups and education levels exhibit varying degrees of proficiency with KMT devices [32, 43]. Therefore, we do not directly explore the relationships between age groups or education levels and KMT dynamics. Instead, we would like to identify which age groups or education levels contribute most to proficiency with KMT devices by using Questions 15 and 16. These two questions do not specify the type of keyboard and mouse.

Question 15 Which one or more age groups are relatively more familiar with using keyboard and mouse in your opinion (multi choice)?

- | | | |
|---|--|--|
| ☐ Under 18 years old | ☐ 18–24 years old | ☐ 25–34 years old |
| ☐ 35–44 years old | ☐ 45–54 years old | ☐ 55–64 years old |
| ☐ 65 years or older | ☐ No difference | ☐ Not clear |

Question 16 Which one or more education levels are relatively more familiar with using keyboard and mouse in your opinion (multi choice)?

- | | | |
|--|--|--|
| ☐ Less than high school | ☐ High school | ☐ Trade school |
| ☐ Bachelor's degree | ☐ Master's degree | ☐ Ph.D. or higher |

The distributions of answers to the questions 15 and 16 are shown by Radar charts in Fig. 6(a) and (b). Each axis in the radar chart represents a response option from the questions, with the value along each axis indicating the number of participants who selected that option. The results show that nearly all participants agree there are differences in the familiarity with KMT devices across age groups and education levels. Generally, individuals aged from 18 to 34 years old are more familiar with these devices, followed by those aged 35 to 44 and those under 18. A broad consensus has been reached that individuals over 44 are less familiar with keyboard and mouse. Similarly, higher education levels tend to correlate with greater familiarity with KMT devices. However, a slight inverse trend appears, as participants with only a bachelor's degree report higher familiarity than those

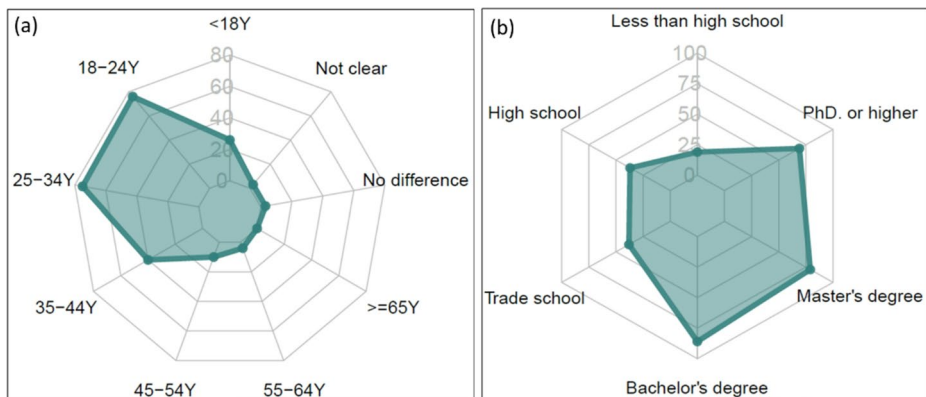


Fig. 6 Distribution of (a) age groups and (b) education levels in terms of familiarity with KMT devices. (Each axis in the radar chart represents a response option from the questions, with the value along each axis indicating the number of participants who selected that option.)

with advanced degrees such as a master's or a PhD. Thus, we can assume that age and education level are both significant user features leading to different KMT dynamics. These findings align with previous research [32, 43], which indicate that people aged from 16 to 29 years old typically have higher typing speed compared with other age groups. A reason for this is that this group of people are more likely to already gain or be pursuing a higher education level, and this in turn requires them to use these input devices more frequently in their daily work and life, making them more proficient with devices.

Table 6 shows that K6, K7, K18, and M19 are significantly influenced by hand using habit. Typing with both hands on a touchscreen device, such as a smartphone or tablet, tends to result in faster input with fewer errors, lighter screen presses, and quicker finger or stylus movements. This is because the left and right margins of the screen are easily accessible to the corresponding hands, and the central area (especially on large screens) can be reached comfortably by fingers from both hands.

Table 7 shows that software proficiency is also a significant user feature influencing F3, K18, M19, F1, and F2. Much software typically requires both keyboard and mouse interaction. Software with rich graphical user interfaces requires more mouse-based interactions for efficient operation. In this case, when users are familiar with keyboard shortcuts and the functions of interface buttons within the software, they tend to achieve higher keystroke speed, mouse movement speed, and mouse click speed, since they do not need to spend time figuring out the required actions or how to perform them. They also tend to use both keyboard and mouse in a more fluid and coordinated manner.

4.3 Quantitative analysis

In the quantitative study, participants' responses on the questions "what are their perceived impact levels" expressed as percentage values, which were used to analyse the overall impact levels of each user feature on KMT dynamics. The impact level for each user feature was presented as a sliding bar ranging from 0 to 50% in the questionnaire, allowing participants to adjust their responses (for example, see Question 14 above). When participants selected "no difference" in the last question, the default value for this sliding question was set to -3%. The larger the number is, the higher the impact level is. A total of 107 responses were recorded for each sliding question, with percentage values ranging from -3 to 50%. The quantitative analysis of these responses provides a general guideline for further study to normalize personalised variations in general emotion recognition.

Figure 7 presents 8 boxes in one plot summarising the overall influences of each user features on KMT dynamics. Each individual box shows a distribution of 107 responses for the corresponding user feature as indicated in the legend. The dark line within each box represents the median value of corresponding 107% data, which marks the mid-point of the data thus half the data are greater than or equal to this value and half are less. The middle "box" represents the middle 50% of data for participants' responses. The range of data from lower to upper quartile is referred to as the inter-quartile range. The middle 50% of data fall within the inter-quartile range. 75% of the data fall below the upper quartile. 25% of data fall below the lower quartile. The upper and lower whiskers represent data outside the middle 50%. The white point denotes the average value of the corresponding data.

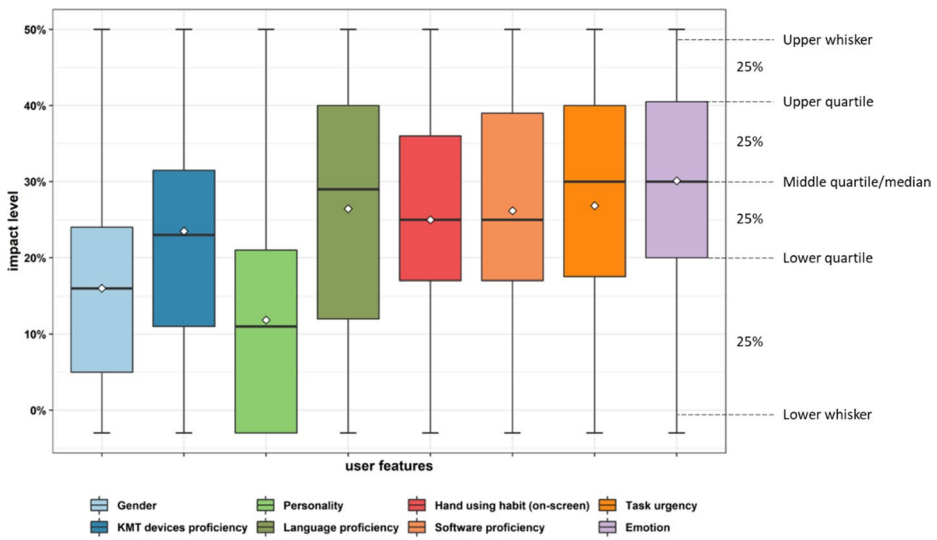


Fig. 7 Box plot describing the distribution of participants perceived impact levels of eight user feature candidates on overall KMT dynamics

The median provides a robust measure of central tendency that is less affected by outliers or skewed data than the average. Thus, this study uses median value as the main measure to interpret the box plot. The equation of calculating median value of a sequence of ordered data is shown in Eq. (3):

$$\text{Median}(X) = \begin{cases} x_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd,} \\ \frac{x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}}{2}, & \text{if } n \text{ is even.} \end{cases} \quad (3)$$

where X is a dataset with data is sorted in ascending order as $X = \{x(1), x(2), x(3), \dots, x(n)\}$, and n is the number of data in X .

Figure 7 shows that emotion, task urgency, and language proficiency have the most significant impact on KMT dynamics, with median values of approximately 30%, 30%, and 29%, respectively. In contrast, participants considered personality to be the least influential user feature, with a median difference level of around 11%. Hand using habit, software proficiency, and KMT device proficiency exhibit similar levels of influence, with median values of 25%, 25%, and 23%, respectively. Meanwhile, gender is perceived as the second least important user feature, with a median difference level of 16%.

These perceived difference levels align with the qualitative analysis. The more significant a user feature is (more participants think it causes differences on more KMT dynamics more significantly), the higher its corresponding difference level.

5 Results and discussions

5.1 Important (shortlisted) user features to be included in a user profile description

We shortlist seven important user features from ten user feature candidates based on the analysis in the previous section, which are shown in Table 8. The seven user features affect seven KMT dynamics features: K6, K7, K18, M19, F1, F2 and F3. Each user feature affects a specific KMT dynamics feature in a different direction (positive (↑), neutral (-), or negative (↓)), and with varying levels of impact represented as the median percentage, listed in the right column of Table 8. For instance, improved KMT proficiency positively affects K6, M19 and F1, negatively impacts on K7, and has no/neutral effect on F2 and F3. But it is not clear whether the influence of better KMT proficiency is positive or negative on K18. The estimated overall impact of KMT proficiency on KMT dynamics is 23%. All the important user features show at least 16% impact on overall KMT dynamics behavior. Especially, Emotion and Task urgency are the top two user features, each contributing a 30% impact level.

5.2 Streamlined KMT dynamics features

Table 8 shows that the total number of KMT dynamics features are seven, and some of them share similar response/impact patterns to each key user feature. Therefore, to reduce the total number of KMT dynamics features that need to be computed and used in the later machine learning process, these KMT features can be streamlined into grouped features. In practical applications, any one feature from the group can be selected based on how easily it can be obtained.

Table 8 Impacts of important user features on KMT dynamics features

Dynamics features		K6	K7	K18	M19	F1	F2	F3	Median percentage of impact level
User features									
Gender	Male	-	-	↑	↑	↑	↑	-	16%
	Female	-	-	↓	↓	↓	↓	-	
KMT proficiency	More familiar	↑	↓	Different	↑	↑	-	-	23%
	Less familiar	↓	↑	Different	↓	↓	-	-	
Language proficiency	More proficient	↑	↓	-	↑	↑	-	-	29%
	Less proficient	↓	↑	-	↓	↓	-	-	
Hand using habit	Two-hand	↑	↓	↓	↑	-	-	-	25%
	Single hand	↓	↑	↑	↓	-	-	-	
Software proficiency	More Familiar	-	-	↑	↑	↑	↑	↑	25%
	Less familiar	-	-	↓	↓	↓	↓	↓	
Task urgency	Urgent task	↑	↑	↑	↑	↑	↑	-	30%
	Normal situation	↓	↓	↓	↓	↓	↓	-	
Emotion	Agitation	↑	↑	↑	↑	↑	↑	-	30%
	Soothing	↓	↓	↓	↓	↓	↓	-	

As an example, both K18 (key/screen press strength) and F2 (mouse clicking strength) describe the momentum force applied to the device. However, availability varies, sometimes both K18 and F2 are accessible and other times only one is available such as K18. In addition, different software emphasizes different input methods. For example, Microsoft Word primarily relies on keyboard input, whereas graphic software like Solidworks involves mostly mouse input. Selecting one of these features is sufficient for a specific task, such as using K18 for Microsoft Word or F2 for Solidworks®. There is no need to use both simultaneously all the time. This rationale supports the streamlining of them into the single grouped feature K18(F2), as shown in Table 9.

This pattern is even more evident in Table 8. For example, K6, M19, F1, and F3 generally show similar response/impact patterns to the same user features, because M19, F1, and F3 describe the speed of a single action when hitting on a key, a mouse button or touchscreen, while K6 reflects average speed over a period of time. They are all about the input speed. The selection of a specific feature can depend on the type of device being used (for example, when using a touchscreen device, M19 will be a suitable choice) or the type of application in use (when using a desktop with Microsoft Word, K6 or F3 will be appropriate, but when working with Solidworks®, F1 will be more relevant). Therefore, we streamline them into the grouped feature M19 (K6, F1, F3), as shown in Table 9.

Within a single group, analysing changes in one feature allows for a qualitative estimation of changes in other related features. The benefits of streamlining some features into a group are two-fold. First, it enables selection of the most accessible and easy-to-measure feature, making data collection more practical. Second, reducing the number of key features results in a shorter feature vector which is more efficient for feeding into machine learning or deep learning algorithms for emotion recognition, and thus lowering the time cost. The streamlined key KMT dynamics feature groups are shown in Table 9.

The selection of KMT dynamics features also depends on the user features being studied and the specific application context. For example, F1 is not applicable for hand using habit and K6 is not suitable for Software proficiency. Different application scenarios fur-

Table 9 Summary of the questionnaire study results: key user features and streamlined KMT dynamics features

Dynamics features		K7	K18 (F2)	M19 (K6, F1, F3)
User features/emotion				
Gender	Male	-	↑	↑
	Female	-	↓	↓
KMT Familiarity	More familiar	↓	Different	↑
	Less familiar	↑	Different	↓
Language proficiency	More proficient	↓	-	↑
	Less proficient	↑	-	↓
Hand using habit	Two-hand	↓	↓	↑
	Single hand	↑	↑	↓
Software proficiency	More familiar	-	↑	↑
	Less familiar	-	↓	↓
Task urgency	Urgent task	↑	↑	↑
	Normal situation	↓	↓	↓
Emotion	Agitation	↑	↑	↑
	Soothing	↓	↓	↓

ther influence the choice of features. Typing tasks, where users primarily interact through the keyboard, are better evaluated using typing speed and key/screen pressure. In contrast, gaming scenarios that involve frequent mouse use benefit more from features such as mouse movement speed (or clicking speed) and clicking strength, as mouse data is more abundant than keyboard data in these cases.

5.3 Key impact relationships between the important user features and key KMT dynamics features

To highlight the final identified important user features and KMT dynamics features, we provide a summary of their impact relationships between each important user feature and each KMT dynamics feature in Fig. 8. This summary offers readers a clear overview of their relationships, including the levels of impact.

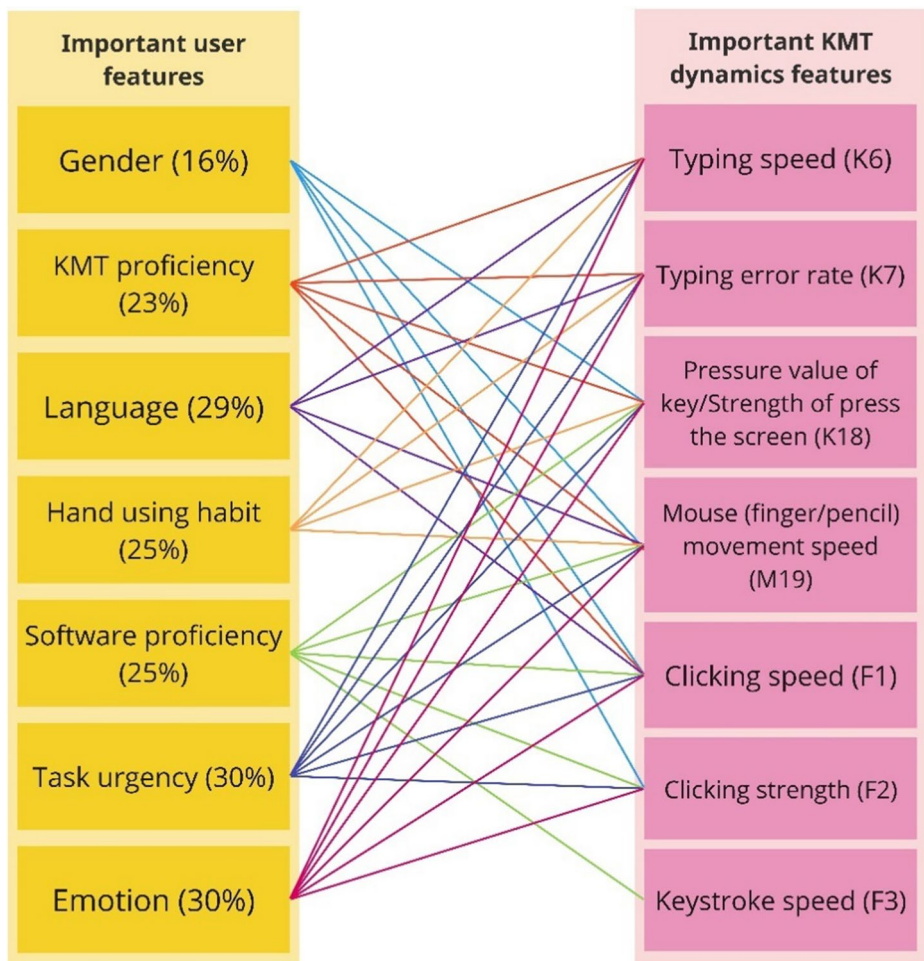


Fig. 8 Key impact relationships between the important user features and key KMT dynamics features

5.4 Personalized emotion recognition framework

The final identified user features and KMT dynamics features were used to refine the initial personalized emotion recognition framework. The updated framework is shown in Fig. 9.

Our personalized emotion recognition framework can provide better guidelines for developing emotion recognition systems based on individual user profile that incorporates multiple important features. Current emotion recognition methods are often too general and less effective when applied across users with differing characteristics. While our framework considers user features, device features, and related behavior features to describe their coupling relationships. This integrates emotion generation and emotion recognition into a single system structure to provide clear guidelines for realizing personalized emotion recognition by better understanding of their coupling relationships.

Understanding which user features and how these user features affect KMT dynamics will help us integrate user factors into emotion recognition according to the relationship framework shown in Fig. 9. Behavior is coupling with stimuli, user features, devices (specifically for KMT devices), and emotions. Giving the same stimuli and device for every user, and after knowing the correlations between user features and behavior, the emotion will be recognized by decoupling the function. And the recognition accuracy would be supposed to be higher because every other relevant user factor has been considered.

Figure 9 provides a framework for personalized emotion recognition by understanding the relationships among the user features and measurable KMT dynamics behavior. This understanding can help filter out influences on measured KMT dynamics (decoupling user

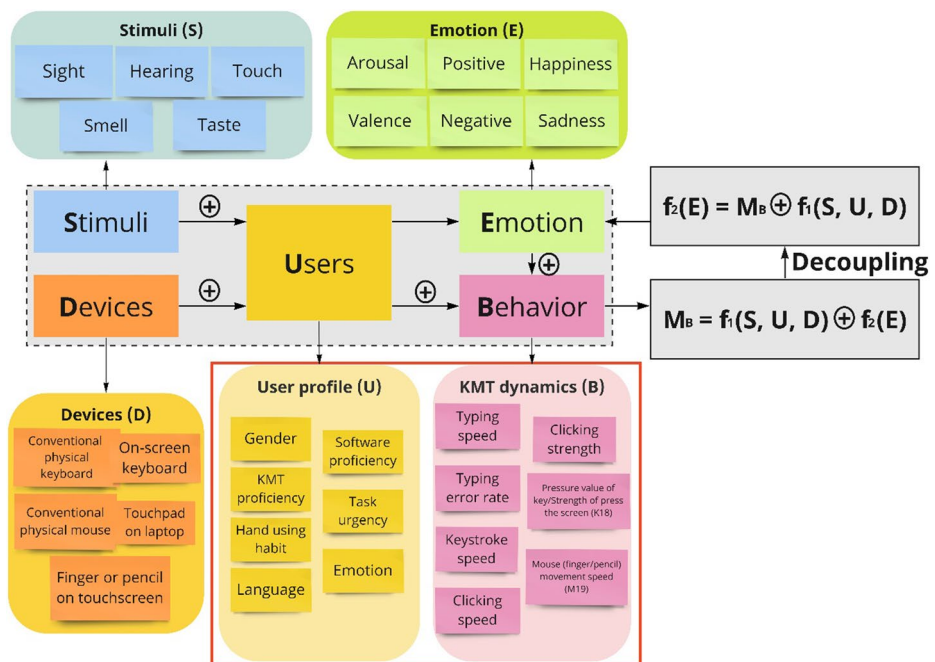


Fig. 9 Refined personalized emotion recognition framework

feature factors from the measurement), and then remained changes in KMT dynamics might better reflect the personalized emotional influence on KMT dynamics, which can lead to a better machine-learning model for emotion recognition from KMT dynamics features.

Based on the qualitative and quantitative findings of this study, we envision a strategy about how to integrate user features with KMT dynamics features to better recognize emotions. Giving the same stimuli and devices, KMT dynamics features and user features will be collected. Then the KMT dynamics features will be adjusted according to different user features. For example, if a user is male, then the values of K18, M19, F1, F2 will be increased totally by 16%. This amount of change should be tuned in some way in the later personalized emotion recognition via a machine-learning. The value of 16% provides a general reference value. For finding a more precise reference value, further quantitative research is needed. For example, how much each user features influence each KMT features is still not clear.

5.5 Limitations

This study applied a questionnaire survey as the main research methodology. This method is suitable to explore the relationships between user features and KMT features in an explainable way, since it enables efficient collection of standardised responses from a diverse participant pool, ensuring consistency and generalisability in data analysis and findings. However, this method may also raise some limitations. First, it has the limited depth of responses. Unlike interviews or focus groups, the questionnaire constrained participants' ability to elaborate on their experiences, potentially overlooking nuanced behavioural factors influencing KMD. The second limitation is the self-reported data bias. Participants' responses are subjective, relying on self-perceptions rather than objective measurements, which may have introduced bias or inaccuracies. For example, everyone has their own understanding about the words like "frequently" and "occasionally". There is no standard definition for each of them, so participants may choose one option just based on their own understanding. The third limitation is the lack of contextual understanding. The questionnaire could not fully capture workplace dynamics, emotional states, or environmental factors influencing KMD, limiting its explanatory power. The fourth limitation is the fixed response options. Predefined questions and response categories may have restricted participants' ability to express unique insights, reducing the richness of qualitative data.

6 Conclusion

This study has investigated how and to what extent a user profile feature can influence one's keyboard, mouse, and touch screen usage in KMT dynamics. To achieve this aim, we first conducted a questionnaire survey with 107 participants in our qualitative study to collect opinion data. Then we performed data analysis to explore the relationships between the initial set of user feature candidates and the initial set of KMT dynamics feature candidates. We then identified what user features are important to KMT dynamics and in turn recognized what KMT dynamics are important for emotion recognition, from their relationship exploration. As a result, we have found seven important user features which are: gender, KMT proficiency, language proficiency, hand using habit, software proficiency, task

urgency, and emotion. The initially identified seven KMT dynamics features are also confirmed as important features for emotion recognition, which are F1, F2, F3, K6, K7, K18, and M19. Second, we conducted quantitative analysis of the survey data, and identified to which extent (degree) one key user feature could sensibly affect important KMT dynamics features. Finally, we updated our emotion recognition framework with identified important user features and KMT dynamics features as a future development guidance model for further research in the related emotion recognition field.

In summary, this study has made the following contributions:

1. Identification of key user features that significantly influence the behavior related to KMT devices usage.
2. Identification of key KMT dynamics features as important behavioral indicators for emotion recognition.
3. Understanding of how these user features affect the KMT dynamics features from both qualitative and quantitative perspectives.
4. A new personalized emotion recognition framework that integrates the identified key user and KMT dynamics features into the emotion recognition pipelines.

There are three directions for future work. The first one is to conduct better qualitative study with more balanced participants' profiles. Also, the sample size could be bigger. The second direction is to consider more task-related features. Currently, only task urgency has been considered. But, in the future, different types of work should be considered. For example, some tasks (hard tasks) require higher mental workload, and some (easy tasks) require lower mental workload, and in the middle, they are normal daily tasks requiring medium level of mental workload. Different tasks require different skills and mental workload; thus, different types of tasks may affect KMT dynamics without considering task urgency. And how combined features of task urgency and task type affect KMT dynamics features is also future research work. And the third direction is to design and develop a digital tool for personalized emotion recognition to quantitatively evaluate how key user features affect KMT behavior and in the reverse direction how to recognize user features such as personality, gender, or software proficiency, by detecting user's emotional states at the same time from KMT dynamics features.

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Author contributions Liying Yang conducted the research and paper writing; Shengfeng Qin conducted research supervision, research planning and proof-reading.

Data availability the survey data is available on request.

Declarations

Ethics approval and consent to participate The research has been approved by Northumbria University Ethics Approval System and all participants have consented to participate.

Competing interests The authors declare that they have no known financial or non-financial competing interests that are directly or indirectly related to the work submitted for publication.

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